SURJO DEY

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EDUCATION

Rajiv Gandhi Institute of Petroleum Technology, Raebareli, India **2022 – 2026**
B.Tech in Computer Science and Engineering *CGPA: 3.22/4.00*
Recipient of the Government of India's Study in India (SII) Scholarship Program

RESEARCH EXPERIENCE

i3 Research Lab, Harvard Medical School, Boston, USA **Nov'26 – Present**
Research Intern — Supervisor: Dr. Yangming Ou *Remote*
Researching and developing medical LLMs for domain specific question answering and automated clinical report understanding using advanced language modeling.

Tezpur University, Assam, India **Mar'26 – Oct'26**
Research Fellow — Supervisor: Dr. Swalpa Kumar Roy *Remote*
Worked on Vision-Language Models (VLMs), build better ways to align image and text representations for stronger visual grounding and understanding across multimodal tasks.

Jashore University of Science and Technology, Jashore, Bangladesh **Jun'25 – Sep'25**
Research Intern — Supervisor: Dr. A F M Shahab Uddin *On-site*
Developed segmentation-guided diffusion models for medical image augmentation using the Duke-Breast-Cancer-MRI dataset to improve robustness and variability in diagnostic model training data.

PUBLICATIONS

A. Das, **S. Dey**, K. Biswas, S. K. Roy, M. Abdar, V. K. Verma. "**SALD: Self-Referenced Advantage Learning for Diffusion Models**". *Submitted:* ***ICLR***, 2027 [Preprint]
Proposed a training framework that improves image generation by comparing predictions at different noise levels using a shared model, without requiring an external teacher network.

S. Dey, P. Saikia. "**Explainability in Generative Medical Diffusion Models: A Faithfulness-Based Analysis on MRI Synthesis**". *3rd World Conference on Smart Computing Systems (WCSC)*, 2026 [Paper]
Proposes a faithfulness-based explainability framework using prototype methods to make generative diffusion models transparent for breast MRI synthesis.

S. Dey, A. Sharma, H. Raj, S. Biswas. "**Automated Detection and Analysis of Minor Deformations in Flat Walls Due to Railway Vibrations Using LiDAR and Machine Learning**". *International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, 2024 [Paper]
Proposes an automated LiDAR and machine learning pipeline to detect and quantify minor structural wall deformations induced by railway vibrations in urban infrastructure.

S. Dey, A. Sharma, H. Raj, S. Biswas. "**Comprehensive Analysis of Structural Defects in Various Structures Using TLS Data and Machine Learning**". *IEEE International Conference on Future Machine Learning and Data Science (FMLDS)*, 2024 [Paper]
Develops an automated ML-based system using TLS point cloud data to detect and analyze structural deformations across various building geometries.

Some other Papers are under review on Core Ranked Conferences and Top Ranked Journals

TECHNICAL SKILLS

Languages	Python, MATLAB, C
ML / DL	PyTorch, Diffusers, Transformers, Scikit-learn, Pandas, NumPy, Matplotlib
Research	Diffusion Models, Vision-Language Models, Fine-tuning, Curriculum Learning
Tools	LaTeX, Git, Jupyter, VS Code, Google Earth Engine
OS	Linux (Ubuntu), Windows

REFERENCES

Dr. Pallabi Saikia
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Dr. Vinay Kumar Verma
Senior Applied Scientist
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Formerly Duke University, USA
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